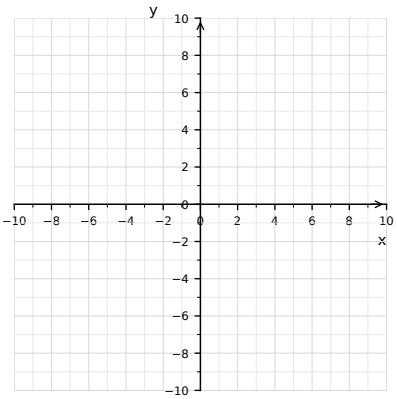
TSM 099 Final Exam

Practice 2 | Version 202610122020.32

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

QUESTIONS	ANSWERS
1. Find the intercepts and graph the line on the grid. $x + 2y = 8$	1. Intercepts 4 pts Graph  6 pts
2. Solve using the quadratic formula. Give exact simplified solutions. $3x^2 = 5 - x$	2. Solutions 10 pts
3. Find the slope-intercept equation through these two points. $(2,\ 1);\ (4,\ 3)$	3. Equation 10 pts
4. Multiply and simplify. $\frac{x^2 - 9}{x^2 - 2x - 48} \times \frac{x + 6}{x^2 + 3x}$	4. Expression 10 pts
5. Add and simplify. $\frac{2}{x^2 - 3x} + \frac{3}{x^2 - 9}$	5. Expression 10 pts
6. Solve by taking square roots. $3(x - 2)^2 = 48$	6. Solutions 10 pts
7. Solve by factoring. $x^2 + 2x = 15$	7. Solutions 10 pts

TSM 099 Final Exam

Practice 2 | Version 202610122020.32

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

QUESTIONS	ANSWERS
8. Solve and check in the original equation. $\frac{10}{2x-8} + \frac{2}{x-4} = \frac{7}{3}$	8. Solution/check 10 pts
9. Factor completely. $10x^2 - 11x - 6$	9. Factors 10 pts
10. Evaluate at x = -2; show substitution. $(x+3)(3x-6)$	10. Value 10 pts

Instructor answer key: TSM 099 Final Exam

Practice 2 | Version 202610122020.32

Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

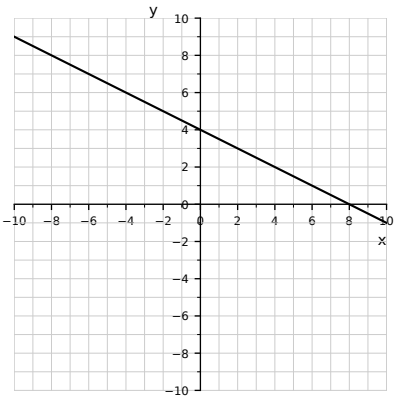
TEST PAGE 1

1. Intercepts

$(8,\ 0);\ (0,\ 4)$

4 pts

Graph



6 pts

2. Solutions

$x = -\frac{1}{6} + \frac{\sqrt{61}}{6},\ \ -\frac{\sqrt{61}}{6} - \frac{1}{6}$

10 pts

3. Equation

$y = x - 1$

10 pts

4. Expression

$\frac{x-3}{x(x-8)}$

10 pts

5. Expression

$\frac{5x+6}{x(x-3)(x+3)}$

10 pts

6. Solutions

$x = -2,\ 6$

10 pts

7. Solutions

$x = -5,\ 3$

10 pts

TEST PAGE 2

8. Solution/check

$x = 7$

10 pts

9. Factors

$(2x-3)(5x+2)$

10 pts

10. Value

-12

10 pts